

Towards a mechanistic understanding of the drivers of plant-pollinator interactions

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Abstract

Flowering plants exhibit an astonishing diversity of reproductive structures, which shapes how they interact with their pollinators. However, to what extent reproductive traits or other drivers like phenology and species abundances shape plant-pollinator interactions is still an open question. To disentangle the relative effects of species phenology, abundances and traits in plant-pollinator interactions, we compared pollinator visitation rates, diet breadth, specialization from plant species that vary in reproductive traits, phenology and abundances. To

do this, we observed plant-pollinator interactions for a whole flowering season in three different botanical gardens located in an urban setting. We anticipate that the ecological role of the different plant and pollinator species are largely driven by their phenological overlap and abundance, and only certain taxonomic groups are strongly influenced by trait matching processes. Our results highlight the relevance of considering multiple mechanisms to understand plant-pollinator interactions.

Introduction

1st paragraph

General introductory paragraph

Ecological networks... Introduce drivers End with limitations and state of the art, temporal and spatial components, sampling biases and data limitations from rarity.

2nd paragraph

Highlight the importance of abundance:

There is consensus that including abundances in our view of ecological processes shaping network patterns yields consistently better results [REFS, eg., annual reviews]. However, most ecological studies...

Highlight the importance of phenology: Highlight the importance of traits/phylogeny:

Last paragraph, introduce what we test!

By conducting this experiment in botanical gardens, we have as objective reducing the “noise” associated with the context-dependent nature of ecological interactions [REF].

Main objective

Explore how traits, phenology and abundances shape species interactions in three botanical gardens from Germany.

Pending tasks:

Question 1:

- Explore relative contribution of ecological drivers (abundances, phenology, traits, phylogeny (plants and bees)).
 - Pollinators: conduct rarefaction on our data and compare phenobs vs random stops vs all
- Associated with this, how increasing sampling impact our view of those processes.

- Temporal grain, how does it affect network patterns. Relative contribution of temporality to network structure or patterns (e.g., day 1, only adds 1 spp versus day 10 that adds 30).

Methods

Study site and sampling method

Plant-pollinator interactions were monitored during the entire flowering season of 2023 in the botanical gardens of Halle, Jena, and Leipzig (Germany). These gardens belong to PhenObs initiative ([Nordt et al., 2021](#)), which consists of a network of botanical gardens across the Northern Hemisphere that records phenological observations on a large set of herbaceous species. The objective of PhenObs is to evaluate the impacts of climate change on species performance (e.g., [Sporbert et al., 2022](#)). In this study, plant-pollinator interactions were recorded for a subset of species from the PhenObs project within these three gardens. Each garden was visited weekly, and plant-pollinator interactions were sampled during 15 mins on each focal plant. Only pollinator individuals that contacted the reproductive organs of the plant were recorded as interacting species. On each sampling day, the different focal plants were sampled in a random order. In addition, to create a comprehensive inventory of pollinators and their interactions within the gardens, 10 random sites were selected each sampling day, and plant-pollinator interactions were monitored for three minutes at each random location. Therefore, two sampling methodologies were implemented, one that recorded interactions on PhenObs plants, and another that consisted of random stops within the garden, with the specific goal of describing all potential pollinators present in the garden.

Sampling Effort and Taxonomic Coverage

This study monitored a total of 78 focal plant species that accounted for x different genera and x different families. In total, we sampled pollinators on 42 plants species in Halle (xmin/species), 43 in Jena (xmin/species) and 47 in Leipzig (xmin/species), and only 12 species were shared across the three gardens.

Data description

To evaluate the different mechanisms that drive plant-pollinator interactions, detailed information on abundance, phenology and species traits for both plants and pollinators was recorded as follows:

- (i) Floral abundance was annotated for each focal plant species, and given that botanical gardens are a semi-control setting with relatively low abundance of pollinators, we were also able to annotate the different number of pollinator individuals on each sampled plant when documenting interactions. Despite not being fully independent of interactions, this approach offers a more realistic estimate of pollinator abundance than the standard method of estimating abundances from interactions, which can bias our understanding of the mechanisms driving network structure (Blüthgen et al., 2006; Vizentin-Bugoni et al., 2014).

Some species had tiny flowers densely packed into compact inflorescences (e.g., Apiaceae, Asteraceae, Dipsacaceae; $N = 53$ species, Table 1), which rendered impractical assessing the number of elemental florets visited by pollinators. In these cases, the number of inflorescences available per patch and visited per census was counted rather than individual flowers, and visitation probabilities (see Data analysis) thus actually refer to inflorescences. For simplicity, I will refer to visitation to both single flowers and inflorescences as “flower visitation.” In some analyses, however, the two types of visitation units will be considered as levels of the discrete variable “visitation unit,” one of the predictors of pollinator visitation considered here (“single flowers” vs. “flower packets”

- (ii) Given that plant phenology was already monitored on the PhenObs project, we have weekly phenological records with flowering percent for all focal plants. For 10 species with missing phenology data, we conducted generative additive models (GAM) with our records of flower number to build the phenology on the same sampling dates as the other species by using the *mgcv* package. Unlike plants, pollinator phenology relies on ‘stochastic’ interaction observations that may not represent their complete flying period. For this reason, we opted to complement our observational data from the Global Biodiversity Information Facility (GBIF). Germany is one of the countries with highest records of occurrence data (Rocha-Ortega et al., 2021), and in particular, their cities (e.g., botanical gardens) contain most of those records (Sweet et al., 2022). Thus, we opted to extract occurrence records around the three cities for the different pollinator species to complement the different pollinator phenologies. We first created a centroid within the 3 cities, and then, created a rectangular shape around it that expanded approximately 1000 kms in longitude and 500 kms in latitude (Figure S1). We used this shape because we expect environmental variables to be more consistent and phenology to be more similar across a longitudinal range than a latitudinal one. Thus, our download resulted in 2,933,772 occurrence records (<https://doi.org/10.15468/dl.k5cnuf>).
- (iii) Plant reproductive traits were selected following the main traits driving the trait spectrum of flowering plants as shown in Lanuza et al. (2023) (i.e., flower number, plant height, style length, flower size, ovule number and selfing). In addition, we also included phenological traits (i.e., flower lifespan and flowering length), floral rewards (nectar volume and number of pollen grains per flower) and seed weight as a measurement of the reproductive investment. Pollinator traits included intertegular distance, body length and proboscis length. Both intertegular distance and body length were measured from

images using ImageJ, and proboscis length was extrapolated for bee species by using the pollimetry package (Kendall et al., 2019). Non-bee species were complemented with proboscis values from the literature and images when possible.

*Phylogeny... Plant phylogeny is easy, bee phylogeny too...

*I think this article may need some sort of conceptual figure

*compute phenology matrices: presence absence and probability ones based by calculating the probability of observing both if they are independent events divided by total number of sampling days

Analysess

Accumulation curves

Compare matrices presence/absence|visitation rate ~ abundance/phenology/traits

Check abundances (FLORAL ABUNDANCES)

Results

Compare differences between phenobs and random stops across gardens (identity, number of species and interactions by sampling date or sampled species). Number of species of plant and pollinators on each and percentage of plant-pollinator interactions and pollinator species captured by the PhenObs species.

*Rarefy by garden: phenobs, random stops and all.

PROVIDE: Percentage of forbidden links due to phenological mismatches by garden.

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Supporting information

Figure S1: Some sort of combination of the code below and the graph of location from the poster